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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAJOR PROJECT PHASE II (BCS786)

on

“Metaheuristic Deep Learning Models for Leukemia Classification and Grading”

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Introduction

Leukemia is a blood cancer that originates in the bone marrow and disrupts normal blood cell production, leading to serious health complications. Traditional diagnosis relies on manual blood smear analysis, which is time-consuming and prone to human error. With increasing leukemia cases, automated diagnostic support systems are essential. This project proposes a PSO-optimised ResNet-18 model to automatically classify leukemia types and grade disease severity using blood smear images, improving accuracy, speed, and diagnostic consistency.



Literature Survey

S.No	Title of Paper	Authors	Year	Method	Gaps
1	PSO-Based Leukemia Classification Using CNN Hyperparameter Optimization	Ju et al.	2024	Used Particle Swarm Optimization to automatically tune CNN settings for better leukemia classification accuracy.	Model is hard to interpret by doctors; black-box nature.
2	Ant Colony Optimization for Feature Selection in Leukemia Subtype Classification	Heng et al.	2023	Employed Ant Colony Optimization to pick important features before classifying leukemia subtypes using deep learning.	Tested only on small datasets; needs validation on larger data.
3	Genetic Algorithm and SVM Hybrid for Leukemia Detection Using Gene Expression Profiles	Kebaili et al.	2022	Combined Genetic Algorithm with Support Vector Machine to analyze gene data for leukemia detection.	Limited to gene data; not tested with medical images.

4	Hybrid Whale Optimization Algorithm and Random Forest for Leukemia Image Classification	Akbar et al.	2022	Used Whale Optimization to select image features and Random Forest for classification of leukemia from images.	Requires testing on more diverse datasets for robustness.
5	Ensemble of Genetic Algorithm and PSO for Leukemia Grading	Shaham et al.	2021	Combined Genetic Algorithm and PSO for better feature selection and classification of leukemia grades.	Computational complexity is high; needs optimization.
6	Metaheuristic-Based Deep Learning Framework for Leukemia Classification	Zhang et al.	2020	Integrated metaheuristic algorithms with deep learning to improve leukemia classification accuracy.	Model training time is long; interpretability issues remain.

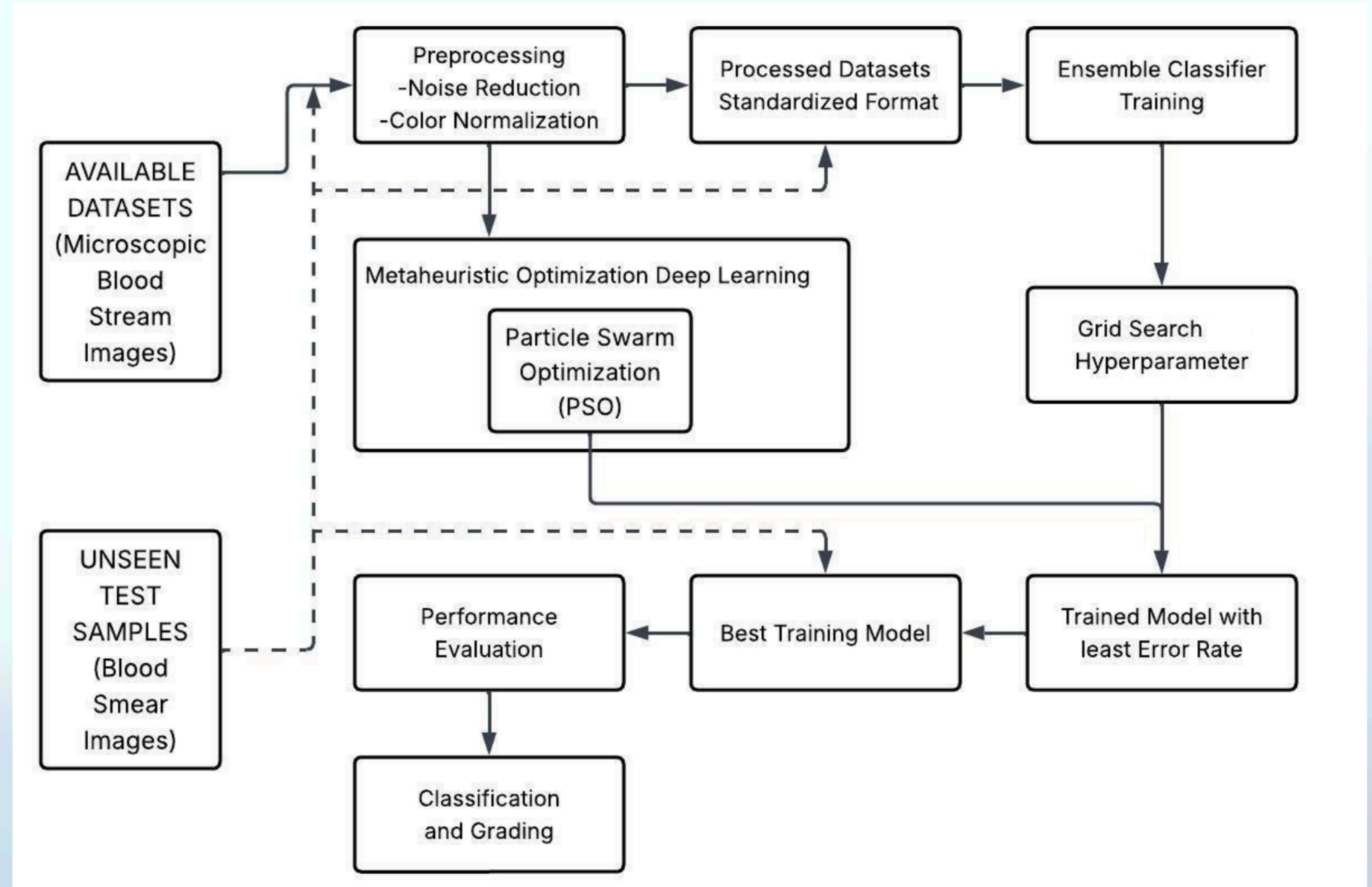
Problem Statement

Leukemia diagnosis requires precise classification and grading to guide effective treatment decisions. Existing automated systems predominantly focus on either classification or grading in isolation, lacking an integrated approach that addresses both simultaneously. Moreover, the accuracy and robustness of these systems are often compromised due to challenges such as imbalanced datasets, high-dimensional feature spaces, and limited interpretability. Current models rarely employ metaheuristic optimization techniques, which can enhance model performance by effectively selecting relevant features and tuning parameters. This gap results in suboptimal diagnostic outcomes and limited clinical applicability. Therefore, there is a critical need to develop a metaheuristic-driven framework that concurrently performs leukemia classification and grading, optimizing diagnostic accuracy and providing reliable, interpretable results suitable for clinical integration.

Project Objectives

- To design and implement a unified system for automated leukemia classification and grading.
- To employ metaheuristic algorithms (e.g., PSO, GA) for optimal feature selection and hyperparameter tuning of classifiers.
- To enhance accuracy, sensitivity, and reliability of classification and grading on imbalanced medical datasets.
- To integrate the classification and grading stages into a single efficient pipeline, enabling streamlined analysis and faster decision support.

Methodology



Methodology

- **Dataset Collection:** Microscopic blood smear images representing different leukemia types are used as input.
- **Image Preprocessing:** Noise removal, colour normalisation, and standardisation to ensure uniform image quality and size.
- **Deep Learning Feature Extraction:** CNN-based models automatically extract relevant features from preprocessed images.
- **Metaheuristic Optimisation (PSO):** Particle Swarm Optimisation tunes model parameters and improves learning efficiency while reducing classification error.
- **Ensemble Training and Hyper-parameter Tuning:** Multiple models are trained for robust prediction, combined with grid search to optimise learning rate, batch size, and network settings.
- **Model Selection:** The best-performing model is chosen based on minimum error and optimal performance.
- **Performance Evaluation:** Evaluation on unseen test data using accuracy, precision, recall, and F1-score.
- **Classification and Grading:** Leukemia classified into ALL, AML, CLL, and CML, with severity graded as Grade-1, Grade-2, and Grade-3.
- **Clinical Decision Support:** The final system provides accurate and consistent leukemia diagnosis and grading for clinical use.

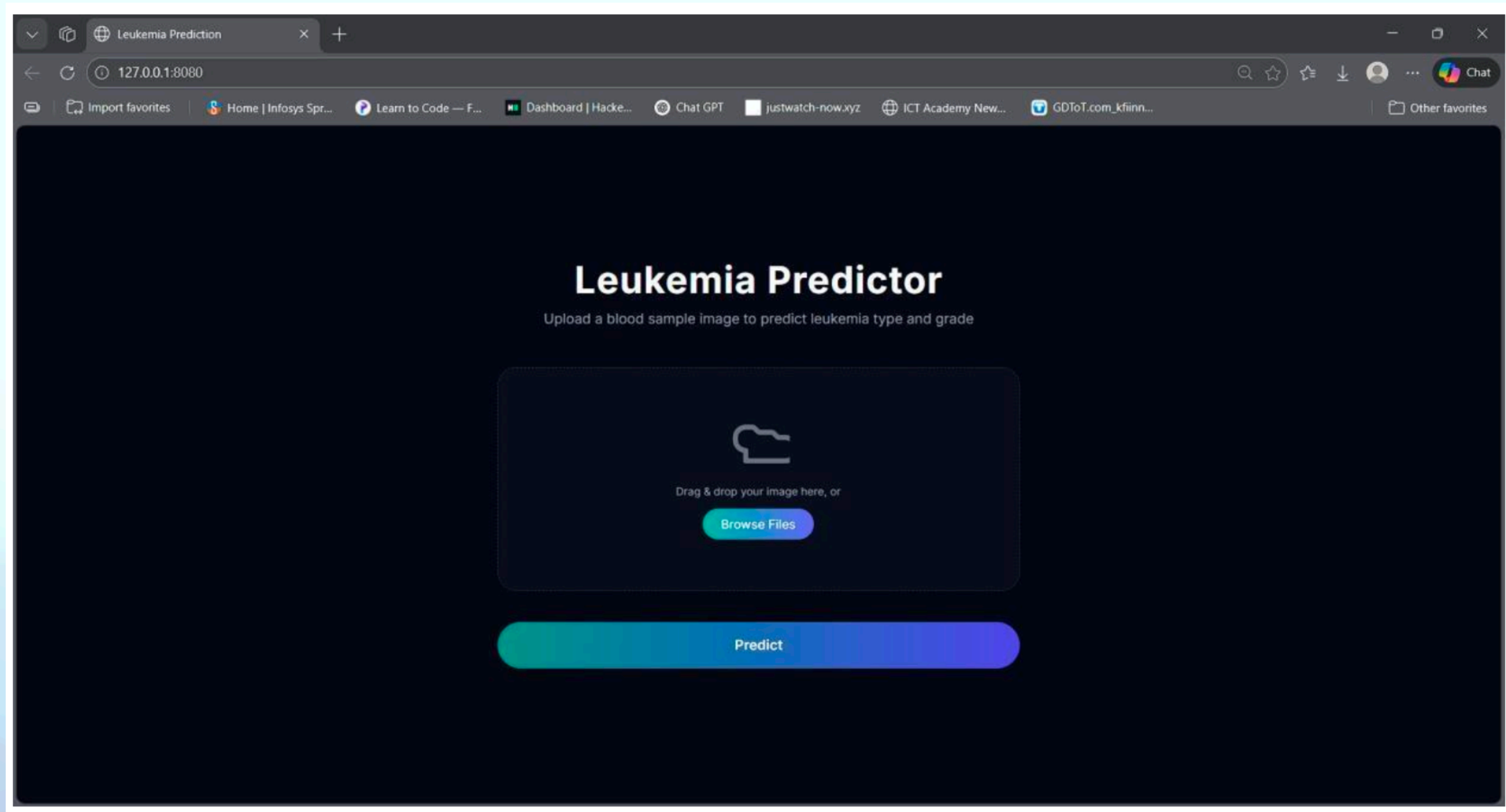
FEATURE PARAMETERS FOR CLASSIFICATION

Feature Parameter	Description	Why It Helps Classification
Nucleus Shape	Round, irregular, folded, multilobed	ALL → round; AML → irregular; CLL → small; CML → granulocyte patterns
N/C Ratio	Ratio of nucleus to cytoplasm	High ratio → acute (ALL/AML); Lower → chronic (CLL/CML)
Chromatin Texture	Fine, coarse, dense	Lymphoid = fine; Myeloid = coarse
Granules	Presence/absence in cytoplasm	Seen in AML/CML; absent in ALL/CLL
Auer Rods	Needle-like structures	Strong indicator of AML
Cell Count Pattern	Number of WBCs/blasts	Acute → many blasts; Chronic → many mature WBCs

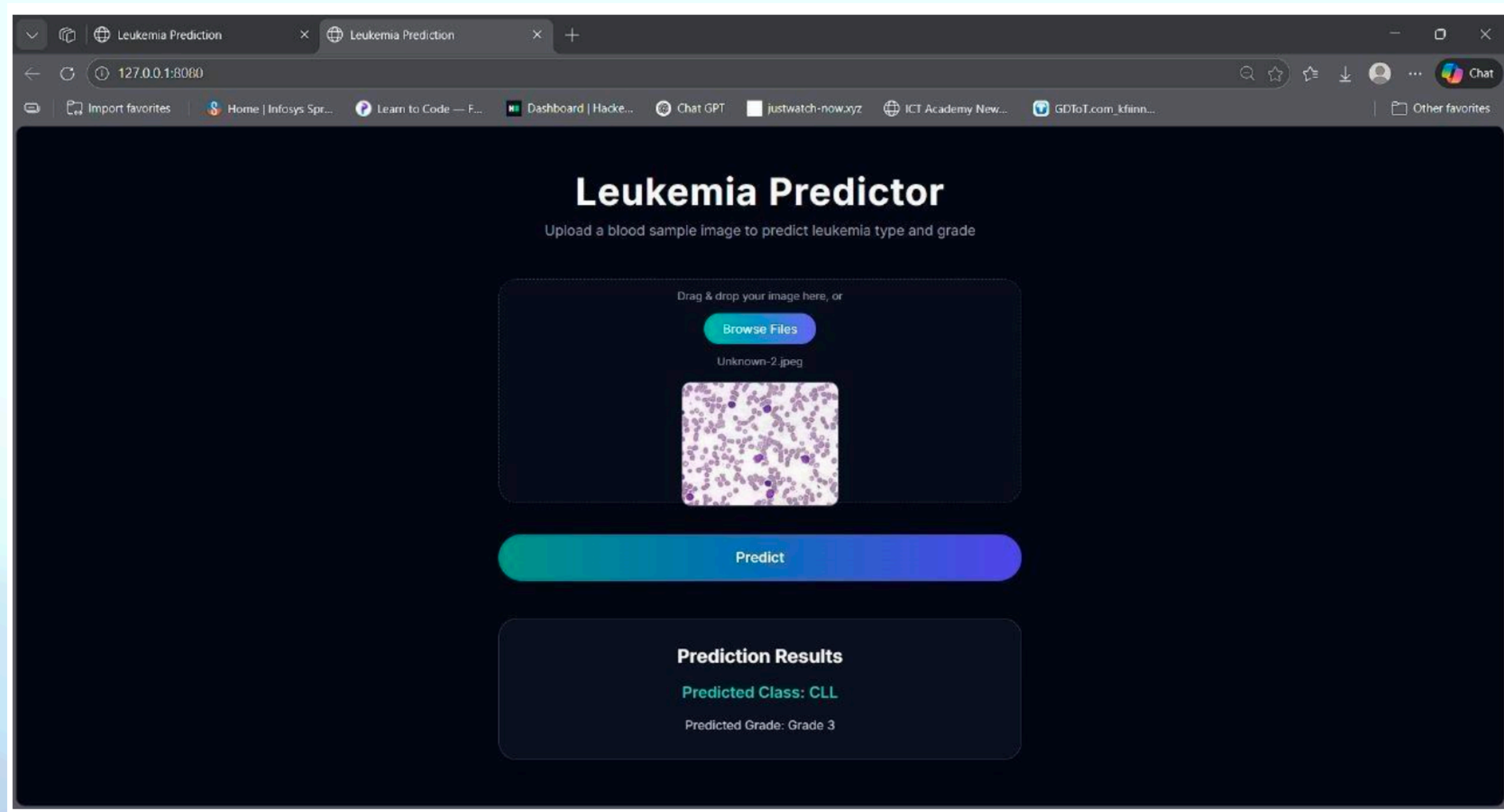
FEATURE PARAMETERS FOR GRADING

Grading Parameter	Description	Grade Indicator
Blast Cell Density	Count of blasts in image	Low = G1, Moderate = G2, High = G3
Morphological Abnormalities	Irregular nucleus, abnormal chromatin	More abnormalities → higher grade
Blast Dominance	% of blasts vs normal cells	High blast dominance = Grade 3
Nuclear Size & Distortion	Enlarged or irregular nuclei	Increased distortion = Grade 3
Background Cell Suppression	Visibility of RBCs/platelets	Less visibility → more severe grade

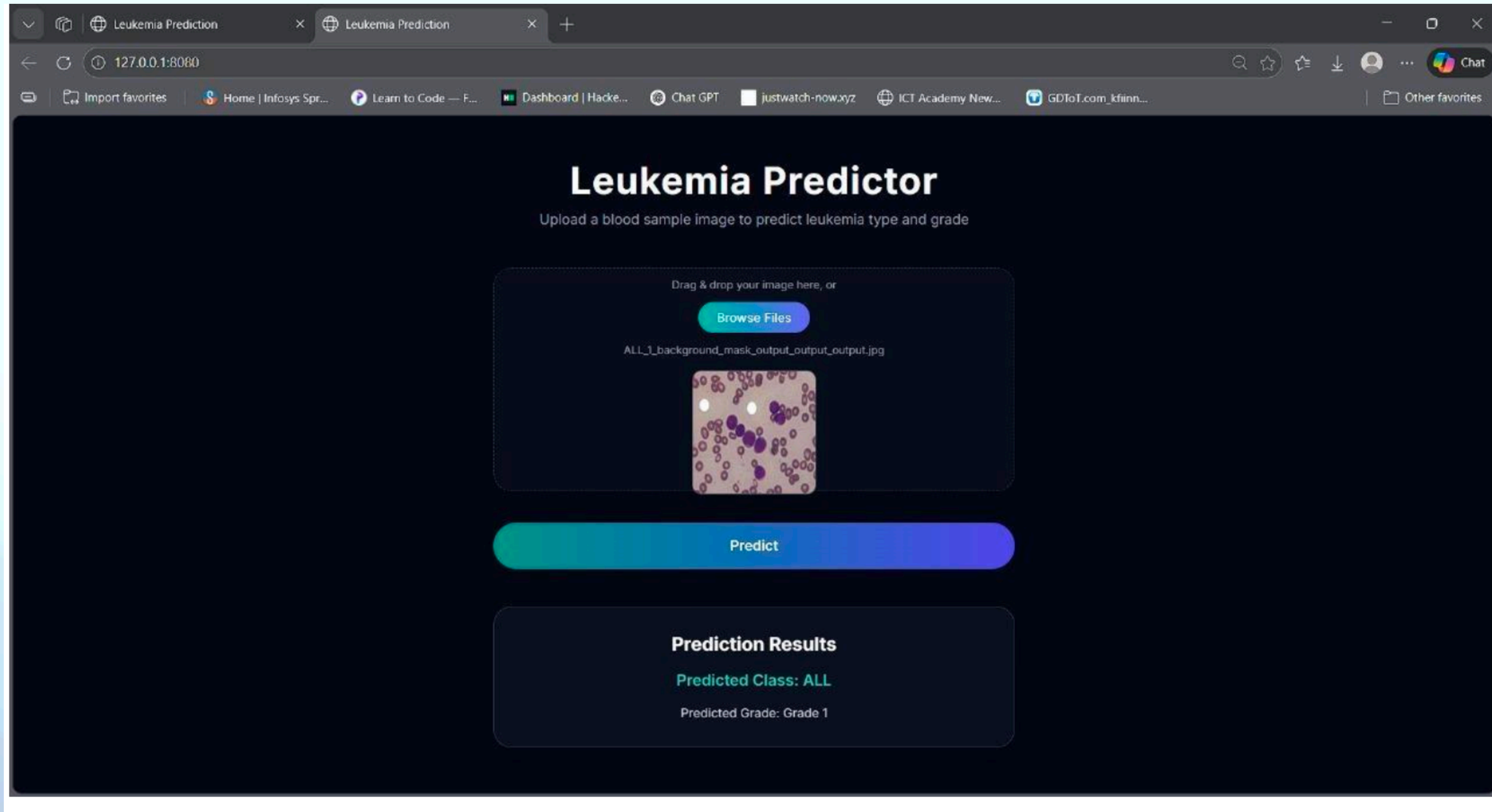
Screenshots:



Screenshots:



Screenshots:



Future Scope...

- **Deeper CNN Architectures:** Use ResNet-34/50/101 to capture complex patterns and improve accuracy.
- **Enhanced Optimization Methods:** Apply advanced or hybrid metaheuristic techniques to boost performance and reduce training time.
- **Multi-Modal Data Integration:** Combine image data with clinical, lab, or genetic information for more reliable diagnosis.
- **Real-Time Clinical Deployment:** Implement the system in hospitals for large-scale, real-time leukemia screening and decision support.

Conclusion



This project presents a PSO–ResNet-18 based framework for accurate leukemia classification and severity grading from blood smear images. Particle Swarm Optimization improves feature relevance and learning configurations, allowing ResNet-18 to focus on critical cellular patterns such as morphology, nucleus structure, and texture. The residual learning capability ensures stable training and strong generalization. Overall, the system offers an efficient and reliable decision-support tool for improved leukemia diagnosis and clinical outcomes.



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